Questions Before Priors:
Causal Dissonance, Insight, and the Origin of Strategic Theories*

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— First draft —

Abstract

The theory-based view treats economic actors as theorists whose causal understandings direct search, experimentation, and action. Existing formal treatments begin after the theory exists. This paper models where strategic theories come from. Agents represent their environment through awareness partitions of a state space; expressible causal theories assign elementary mechanisms to awareness cells; theories and beliefs determine committed policies; and the policy profile causes everyone's consequences through an objective causal system. Bayesian learning operates only within the current language, and a misspecified posterior grows confident rather than alarmed, so detecting representational failure requires a distinct faculty: causal dissonance, the statistical rejection of every expressible theory. Dissonance directs inquiry without determining its output; a costly, fallible technology may produce a previously inexpressible distinction; and a bridge prior restores Bayesian competence on the expanded language. The value of a question is computable before its answers are conceivable, warrant is distinct from post-insight confidence—the insight itself is not evidence—and an agent generally cannot measure its own overconfidence. Because theories cause actions and actions generate evidence, inquiry dies out almost surely, leaving every agent in a Bayesian world; the configurations that remain can preserve false theories, heterogeneous awareness, and permanent performance differences whenever neither strategic variety nor environmental harshness exposes them.

Keywords: theory-based view; unawareness; insight; causal learning; subjective rationality; self-confirming equilibrium

*Preliminary draft; comments welcome. This paper develops the formal core of a research program on pre-Bayesian theory formation in strategy.

1 Introduction

The theory-based view of strategy holds that economic actors are theorists: managers and entrepreneurs carry causal accounts of how value is created, and those accounts—not merely their information or their incentives—direct what they notice, which experiments they run, and what they do (Felin and Zenger, 2017; Felin et al., 2024). A decade of work has established the consequences of holding a theory. Theories guide experimentation and search (Camuffo et al., 2020; Chavda et al., 2024), support the acquisition of resources and rents (Ehrig and Zenger, 2024), and discipline entrepreneurial decision-making in the field (Camuffo et al., 2024). What this literature has not yet formalized is the step before all of that: where a genuinely new theory comes from. Existing formal treatments begin with the theory present—as a menu of candidate theories to select among (Camuffo et al., 2024), as a given causal conjecture that directs search (Chavda et al., 2024), or as a given awareness advantage to be exploited competitively (Ehrig and Zenger, 2024). The origin of the theory is described verbally and documented empirically (Ott and Hannah, 2024), and the absence of formal machinery for it has been explicitly noted as the field's open analytical problem (Felin et al., 2024).

The difficulty is not a missing application of Bayesian reasoning; it is a boundary of Bayesian reasoning. A Bayesian agent updates beliefs over a space of represented possibilities. Conditioning reweights that space and cannot enlarge it, so no sequence of updates makes an unrepresented causal distinction available for belief or action. Worse, when every represented theory is false, the posterior does not sound an alarm: by the logic of misspecified Bayesian learning (Berk, 1966), it concentrates on the least wrong of the available theories, and the agent grows more confident, not less. Within-language updating is self-sealing. The economics of growing awareness has developed axiomatic accounts of how beliefs should extend when the state space expands (Karni and Vierø, 2013, 2017; Vierø, 2021), but in those models the expansion itself arrives exogenously; models of discovery in games describe how play can reveal objectively available actions (Schipper, 2021) without explaining how experience generates a substantive causal question or its answer. What is missing, in both the strategy literature and the economics of unawareness, is a model of the transition: from a broken causal language, through a directed question, to a new representation, and back to ordinary Bayesian life.

This paper models that transition, embedded in strategic interaction. Its organizing object is the chain

$$\text{dissonance} \longrightarrow \text{inquiry} \longrightarrow \text{refined awareness} \longrightarrow \text{new models} \longrightarrow \text{Bayesian judgment},$$

the process by which an agent whose causal representation has failed regains the capacity to be a Bayesian. The back end of the model follows the subjective-rationality tradition (Kalai and Lehrer, 1993b; Ryall, 2003): a finite set of agents adopt causal theories, combine them with beliefs and conjectures to choose committed policies, and the policy profile determines every agent's consequences through an objective causal system that absorbs all market and environmental detail. The front end is new. Agents represent the environment through awareness partitions of a state space; the theories they can formulate assign elementary causal mechanisms to awareness cells; and the partition itself—the language—is endogenous. An awareness cell is an implicit claim that the states it pools are causally alike. Strategic experience can refute that claim: when the statistical record rejects every expressible theory, the agent faces *causal dissonance*, a determinate defect with no expressible repair. Dissonance makes a question salient without providing its answer. A costly, fallible, truth-blind inquiry technology may then produce a refinement of the partition—the formal footprint of an insight—after which a bridge prior restores a probability space and Bayesian judgment resumes on the expanded language.

Five sets of results give the chain its content. First, boundary results locate exactly where Bayesian competence stops: conditioning cannot expand the language, and the misspecified posterior is self-sealing, so detecting representational failure requires a faculty formally distinct from updating. Second, dissonance directs inquiry without determining it. The record restricts admissible repairs—every restorative refinement must separate the evidence clusters that no single mechanism can accommodate—but generally many minimal repairs remain, observationally equivalent worlds demand different answers, and no truth-blind technology can be uniformly correct. Fallibility is the price of a representation that economizes on distinctions. Third, the value of a question is computable before its answers are conceivable. A leverage statistic built from retained experience—invariant across every representation that could answer the question—prices inquiry at exactly the moment the posterior is undefined, and it tracks whether an answer could change action, not how much uncertainty remains. Questions die rationally: sustained success extinguishes them, and defeat concentrates both the evidence and the incentive to inquire in the losing firm. Fourth, the return to Bayes is disciplined by a warrant account: because any successful repair fits the experience that generated it by construction, retrospective fit carries no information, and—when inquiry is truth-blind—neither does the arrival of the insight itself. Post-insight confidence therefore exceeds warrant by a computable gap, and an agent holding a minimal repair generally cannot even formulate the comparison that would measure its own gap. Fifth, because theories cause actions and actions generate the evidence, the population dynamics close the loop. Almost surely, awareness change and inquiry die out, leaving every agent permanently inhabiting a Bayesian world—models fit, questions are dead, inquiry is priced below cost. We call the configurations in which the process rests *self-confirming unawareness equilibria*, and their interest is epistemic rather than strategic: such a configuration can preserve false theories, heterogeneous awareness, and permanent performance differences, and it does so precisely when neither of the two corrective forces—strategic variety inside an agent's categories, or an environment harsh enough to punish misservice observably—is present.

The persistence results deserve emphasis because of what carries them. Performance differences persist in this model not because an equilibrium concept freezes them but because the process that would erase them has died economically: the disadvantaged configuration generates no dissonance, or generates no leverage, or prices its question below the cost of asking. Strategic variety is therefore traced to its source. Agents with identical preferences, identical data-processing competence, and identical objective opportunities end up with different causal languages, different theories, and different payoffs because their experiences posed different questions, their technologies produced different repairs, and their subsequent positions extinguished further inquiry at different points. This is a microfoundation for heterogeneity that begins before probability does.

A deliberately small example runs through the paper—two firms, four customer states, two product architectures—in which every object can be computed exactly: the four minimal repairs of a broken language, the two that survive a simplicity filter, the impossibility of choosing between them from the data, the exact price of the question, the exact warrant gap, and the knife-edge condition under which a market's shared false theory protects itself forever. The example is a laboratory, not the theory; each of its results instantiates a general statement.

The paper proceeds as follows. Section 2 locates the contribution. Section 3 presents the model. Section 4 develops the Bayesian layer and its boundary; section 5 defines dissonance and exposure; section 6 analyzes restoration, direction, and fallibility; section 7 prices inquiry; section 8 develops formulation, warrant, and the return to Bayesian competence; and section 9 closes the loop with the population dynamics. Section 10 discusses implications and the research program.

2 Related literature

The theory-based view. The foundational claim is that theories do more than summarize evidence: they direct attention toward causal possibilities, support counterfactual reasoning, and guide experimentation (Felin and Zenger, 2017). The *Strategy Science* collection on theory-based decisions substantially developed the view (Felin et al., 2024): Camuffo et al. (2024) formalize the choice of which theory to test and show that the value of experimentation grows with the number and uncertainty of candidate theories; Chavda et al. (2024) model search guided by a given causal theory; Ehrig and Zenger (2024) translate theories into awareness structures and show how partial disclosure secures resources and rents; Sorenson (2024) and Adner and Levinthal (2024) probe the mechanisms and the limits of theory-guided experimentation; and Ott and Hannah (2024) document empirically how entrepreneurs develop a poorly formed thesis into an articulated causal model. A parallel empirical program shows that training entrepreneurs to formulate and test theories improves performance (Camuffo et al., 2020). Related conceptual work presses the same open question this paper addresses: Zellweger and Zenger (2023) treat entrepreneurs as scientists generating beliefs under uncertainty, and Felin and Holweg (2024) argue that forward-looking causal theorizing is precisely what data-driven prediction cannot supply. Throughout, the formal treatments condition on the theory, the attribute list, or the awareness advantage being present. This paper supplies the front end those models presuppose, and its equilibrium analysis shows when the supplied front end matters: heterogeneous causal languages, and the performance differences they support, persist exactly when the strategic and environmental conditions identified below fail to erase them.

Unawareness and growing awareness. Standard state-space models cannot represent nontrivial unawareness (Dekel et al., 1998); generalized structures represent multiple awareness levels and interactive beliefs (Heifetz et al., 2006, 2013). On belief revision under expansion, reverse Bayesianism preserves relative odds as the state space grows (Karni and Vierø, 2013), with extensions to awareness of unawareness (Karni and Vierø, 2017), intertemporal settings (Vierø, 2021), unforeseen evidence (Piermont, 2021), and learning under incomplete awareness (Grant et al., 2022). Schipper (2021) distinguishes learning, which discards conceived possibilities, from discovery, which adds unconceived ones, and shows how play can move players to richer games; Schipper (2025) shows that beliefs about novelty can be held without representing the novel objects—a precedent for the delabeled constructs used here. Relative to this literature the paper endogenizes what those models take as given: the expansion event itself. Experience generates the defect, the defect localizes the question, a fallible technology produces the new distinction, and the bridge prior problem—which reverse Bayesianism constrains but does not solve—is embedded in an explicit account of what the generating evidence can and cannot warrant. In strategy, Bryan et al. (2022) bring unawareness into value capture and show that awareness itself earns rents; the present paper models where the awareness advantage originates, which that paper leaves open.

Causal learning with a fixed vocabulary. Causal graphical models separate intervention from association (Pearl, 2009; Spirtes et al., 2000); Ryall (2009) brings this machinery to strategy, modeling imitation as Bayesian inference over operating structures in which passive observation narrows candidates to an observational-equivalence class and experimentation may discriminate among survivors. Active-learning methods select interventions to split a represented equivalence class (He and Geng, 2008). These models are the benchmark for the paper's second learning tier: inference over structures expressible in a fixed vocabulary. The paper begins where they stop—when no expressible structure is adequate—and its nonidentification results extend the equivalence-class logic to the choice among vocabulary repairs: observationally equivalent worlds can demand different distinctions, so no truth-blind repair rule is uniformly correct.

Subjective rationality and self-confirming equilibrium. The back end follows Kalai and Lehrer (1993b) and Fudenberg and Levine (1993): beliefs correct on the path of play need not be correct off it, and subjectively optimal behavior against such beliefs can persist. Ryall (2003) showed that competitive advantage can rest entirely on rivals' self-confirming false theories. Those models fix the hypothesis space; convergence results in the rational-learning tradition require a grain of truth—the truth absolutely continuous with respect to beliefs (Kalai and Lehrer, 1993a). Under unawareness the grain-of-truth condition can fail at the level of language: the truth may be inexpressible. This paper supplies the missing dynamics. Statistical dissonance is the detector of language-level failure, awareness refinement restores the grain of truth on the path of play, and the fixed-language models are recovered exactly as the special case in which awareness never changes. The paper's absorbing configurations extend self-confirming equilibrium with two epistemic conditions—the language survives its own evidence, and no question is worth its price—and it is these conditions, not the strategic ones, that carry the new results.

Statement of contribution. Existing models show how theories guide learning, how Bayesian agents discriminate among represented causal structures, and how play can reveal previously unavailable actions. This paper models a different transition. An agent's causal language pools objectively different situations; strategic experience endogenously reveals an action-relevant failure of that pooling and thereby creates a localized open question; costly inquiry activates a fallible technology whose output is not measurable in the agent's prior awareness; only after that output is formulated does the agent construct a prior and undertake Bayesian judgment, under an explicit account of how much confidence the generating experience licenses. Because actions depend on theories, play can either create the contrasts that generate questions or suppress them—so the model also delivers the population-level answer: when theory heterogeneity and performance differences persist, and when the market erases them.

3 The model

3.1 Agents, states, actions, and consequences

There are n agents, $N = \{1, \ldots, n\}$. Time is divided into generations $g = 1, 2, \ldots$, each consisting of T episodes. In each episode a state $s \in S$, with S finite, is drawn independently from a full-support distribution $\rho \in \Delta(S)$; each agent takes an action a_i from a finite set A_i through a committed policy described below; and a consequence profile $x = (x_1, \ldots, x_n)$ is realized in a finite space $X = \prod_i X_i$. Agent i's consequence x_i is everything the episode delivers to it—profit, sales, observed reactions—and its payoff is a function $u_i : X_i \to \mathbb{R}$ of that consequence. Write $A = \prod_i A_i$ for action profiles.

States are the analyst's descriptions of the payoff-relevant situations agents may face: customer configurations, input conditions, competitive circumstances. Their status in the agents' own representations is the subject of section 3.3.

3.2 Mechanisms and the objective causal system

Definition 3.1 (Elementary mechanisms and the objective system)**.** *A finite set* Θ *of* elementary causal mechanisms *is given, each a map* $\theta : A \to \Delta(X)$*. The* objective assignment *is a function* $\tau : S \to \Theta$*, and the* objective causal system *is*

$$F(\cdot \mid s, a) = \tau(s)(a) \in \Delta(X). \tag{1}$$

Two states are causally equivalent *if they carry the same mechanism;* $\mathcal{P}^\tau$ *denotes the partition of* S *into causal-equivalence classes, and* $\operatorname{marg}_i \theta(a)$ *the* X_i*-marginal of* $\theta(a)$*.*

The causal reading of equation (1) is maintained throughout and does real work. F is a response function, not a statistical summary: it specifies the distribution of consequences that *would* result from every action profile at every state, including profiles never played. Agents' actions, together with the state, cause their consequences. A mechanism is the reduced form of whatever detailed model—a structural causal model of customer behavior, a market microstructure, a production process—underlies that relation; the paper imposes no structural-causal-model formalism, but examples may exhibit micro-foundations explicitly, as the running example's customer market does. Three consequences of the causal reading recur. First, theories, defined next, are interventional objects: they make predictions at unplayed profiles. Second, on-path data constrain F only at visited state–profile pairs, so observationally equivalent systems with different counterfactual content are generic; that is the engine of the identification results. Third, the on-path distribution of consequences given an agent's category of states is a *selection* object—it mixes mechanisms according to which states co-occur with which play—and must never be conflated with any mechanism's own conditional.

3.3 Awareness, theories, and conjectures

Definition 3.2 (Awareness and expressible theories)**.** *Agent* i*'s awareness partition is* $\mathcal{P}_i \in \Pi(S)$*, the set of partitions of* S*. Every consciously available object—theory, belief, conjecture, policy, question—must be measurable with respect to* $\mathcal{P}_i$*. The* expressible theories *are the cell-constant mechanism assignments*

$$M(\mathcal{P}_i) = \{m : \mathcal{P}_i \to \Theta\}, \tag{2}$$

with m *predicting* $x_i \sim \operatorname{marg}_i m(C)(a)$ *at cell* C *under profile* a*. Partition* $\mathcal{P}$ *is* sufficient *for an assignment* τ' *if* τ' *is constant on every cell of* $\mathcal{P}$*, equivalently if* $\mathcal{P}$ *refines* $\mathcal{P}^{\tau'}$*; the objective assignment is expressible for agent* i *if and only if* $\mathcal{P}_i$ *is sufficient for* τ*.*

An awareness cell is thus an implicit causal claim: the states it pools are alike in how actions produce consequences. An expressible theory makes the claim explicit by naming the mechanism. Restricting theories to *single* mechanism assignments—rather than arbitrary conditional distributions per cell—is substantive and deliberate. If any distribution were expressible, any within-cell mixture would be too, and no experience could reject the class: coarseness alone cannot produce a detectable failure in a saturated model space. The mechanism vocabulary Θ is known to all agents from the outset; what an agent may lack is the *distinction* that separates states carrying different mechanisms. Everything the paper calls insight is the production of such a distinction. Expansion of the mechanism vocabulary itself is a deeper dimension of awareness growth deliberately left outside this paper.

Agent i holds a belief $\mu_i \in \Delta(M(\mathcal{P}_i))$ over expressible theories—formally, the acting tier of a two-tier belief specified in section 5—and a conjecture $q_i : \mathcal{P}_i \to \Delta(A_{-i})$ about rival play, both $\mathcal{P}_i$-measurable. Theories and conjectures are different kinds of objects and are deliberately separated: a theory is a causal claim about the objective system; a conjecture is a strategic claim about what other agents will do. The diagnostic apparatus developed below tests theories, not conjectures.

Assumption 3.3 (Observation and retention)**.** *At the end of each episode, agent i observes $(C_{\mathcal{P}_i}(s), a, x_i)$: its awareness cell containing the realized state, the realized action profile, and its own consequence. Episodes are retained as tokens in a deliberative record r_i and can be retrospectively recoded after the agent acquires a new distinction; before acquisition, neither belief nor action can condition on that distinction. The analyst-level history additionally records s itself, so that a newly produced representation can recode the tokens.*

Retention without recodability-in-advance is the minimal separation between experiencing a particular and possessing a projectible concept for it: the agent can remember "customers like that one" without commanding a predicate that distinguishes them, and the technology of section 6 operates on exactly this material. Observability of the full action profile keeps the separation of causal from strategic error clean; a variant folding components of a into x_i changes bookkeeping only.

Assumption 3.4 (Commitment and subjective optimality)**.** *At the start of each generation, agent i commits to a policy $\sigma_i : \mathcal{P}_i \to A_i$ satisfying*

$$\sigma_i(C) \in \arg\max_{a_i \in A_i} \sum_{m \in M(\mathcal{P}_i)} \mu_i(m) \sum_{a_{-i} \in A_{-i}} q_i(a_{-i} \mid C)\, \mathbb{E}_{x \sim m(C)(a_i, a_{-i})}[u_i(x_i)]\,, \tag{3}$$

holds it for the generation's T episodes, and revises beliefs, conjectures, and policies only between generations. When dissonance (section 5) has made μ_i undefined and no new representation has been produced, the previously committed policy is carried forward. Agents evaluate one generation ahead; no discounting is imposed.

Commitment is where statistics and economics meet: a test needs a stable data-generating configuration, and a generation of committed play supplies exactly that. What a deterministic model must assume as a batching device is, in the stochastic model, the natural granularity of learning.

3.4 On-path conditionals

Fix a generation with committed profile $\sigma = (\sigma_1, \ldots, \sigma_n)$. Each state s receives the deterministic profile $a(s) = (\sigma_j(C_{\mathcal{P}_j}(s)))_{j \in N}$. Agent i's *on-path conditional* at an observation pair (C, a) with $S(C, a) = \{s \in C : a(s) = a\} \neq \varnothing$ is

$$p_i^*(\cdot \mid C, a) = \sum_{s \in S(C,a)} \rho\big(s \mid S(C, a)\big) \ \mathrm{marg}_i\, \tau(s)(a). \tag{4}$$

This is the selection object flagged above: rivals with finer partitions condition on distinctions agent i cannot see, so the within-cell state mixture can vary with the profile. It is precisely this dependence—differentiated play by others inside one's own categories—that generates the contrasts on which the diagnostic apparatus feeds.

3.5 Modeling commitments

Five disciplines govern everything that follows; they are what distinguish a model of unawareness from a model of inattention or model selection. (i) Agents hold no prior over the outputs of the inquiry technology and never optimize over named future theories; pre-insight deliberation quantifies only over represented objects. (ii) Questions are generated by defects in the current representation, not drawn from a primitive list. (iii) Inquiry can fail, and its successes can be false; nothing reveals the objective assignment by construction. (iv) All valuations attached to inquiry are subjective, coarse, and possibly wrong; no objective payoff information leaks from behind the awareness frontier. (v) Evidence that generates a candidate representation is separated, in the formal accounting, from evidence that warrants it.

3.6 The running example

Example 1 (A four-state laboratory)**.** Two firms serve a unit market. States are customer configurations $S = \{s_{00}, s_{01}, s_{10}, s_{11}\}$, displayed as a 2×2 array with row coordinate r and column coordinate c; ρ is uniform. Actions are product architectures $A_i = \{0, 1\}$. The mechanism vocabulary is $\Theta = \{\theta^0, \theta^1, \theta^n\}$, with a demand parameter $\beta \in [0, 1]$: under θ^a ("architecture a is preferred"), if the firms differentiate, the firm offering a receives payoff 1 and its rival 0; if they pool on a, each receives $\frac{1}{2}$; if they pool on the other architecture, each receives $\beta/2$, the customer withholding $1 - \beta$ of the unit value. Under θ^n ("architecture is neutral"), each firm receives $\frac{1}{2}$ regardless. A consequence x_i is the pair (own payoff, observed action profile). The objective assignment is the *row law* $\tau(s_{rc}) = \theta^r$: the row coordinate is causally relevant, the column coordinate is not. Both firms begin with the coarse partition $\mathcal{P}_i^0 = \{S\}$, under which the expressible theories are the three constant assignments; the row law is not among them, because it is not constant on S. The baseline sets $\beta = 1$ (deterministic winner-take-all with market splitting); $\beta < 1$ returns in section 9. All mechanisms here are deterministic, so every test below operates in its exact, zero-tolerance instance; nothing in the general development relies on that.

4 Bayesian competence and its boundary

Within fixed awareness, learning is ordinary. Observations representable in the current language update μ_i by Bayes' rule; conjectures are updated empirically from observed play. If no agent's awareness ever changes, the model is a repeated interaction among subjectively rational Bayesian learners in the sense of Kalai and Lehrer (1993b) and Ryall (2003), and its stable configurations are self-confirming equilibria (Fudenberg and Levine, 1993). Everything new in this paper happens at the boundary of that layer, which two results locate.

Proposition 4.1 (No Bayesian emergence)**.** *Bayesian conditioning on any event measurable with respect to the algebra generated by $\mathcal{P}_i$ yields a posterior on $\Delta(M(\mathcal{P}_i))$. No sequence of such updates makes a theory outside $M(\mathcal{P}_i)$ expressible or a distinction finer than $\mathcal{P}_i$ usable in a policy.*

Proof. Conditioning reweights a distribution on its domain and cannot enlarge the domain; every element of $M(\mathcal{P}_i)$ is cell-constant, so every posterior mixture, and every policy optimal against one, is $\mathcal{P}_i$-measurable. $\square$

Remark 4.2 (The posterior is self-sealing)**.** *The stochastic setting yields a sharper point. Suppose every expressible theory is false: the agent's cell mixes mechanisms, so no cell-constant assignment matches the on-path conditionals. A full-support posterior does not signal this. By the logic of misspecified Bayesian learning (Berk, 1966), it concentrates on the expressible theories closest to the on-path conditionals in Kullback–Leibler divergence, and the agent becomes increasingly* confident *in a wrong theory. Posterior mass always sums to one; the inadequacy of the whole model class is not an event in the agent's algebra. Detecting that the language itself is broken therefore requires a faculty outside the posterior—the consistency test of the next section. The distinction matters for how the theory-based view should treat "updating": ranking the answers one can express and judging whether any expressible answer is adequate are different cognitive operations, and only the first is Bayesian.*

Example 1 (continued). Under the coarse partition, firm i's expressible theories are the constant assignments m^0, m^1, m^n. The row law τ is inexpressible: no reweighting of $\{m^0, m^1, m^n\}$, on any evidence, produces it (proposition 4.1). A firm that will later observe payoffs irreconcilable with all three theories will find its posterior undefined, not alarmed—the deterministic instance of remark 4.2, in which misspecification announces itself only because likelihoods hit zero exactly. With stochastic mechanisms, no announcement would occur at all.

5 Causal dissonance

5.1 Tests and active sets

Definition 5.1 (Consistency test and active set)**.** *Fix a tolerance $\eta \geq 0$, a minimum count $\bar{n} \geq 1$, and a nonincreasing slack sequence $w(\cdot) \to 0$. Given record r_i with counts $n_i(C, a)$ and empirical conditionals $\hat{p}_i(\cdot \mid C, a)$, theory $m \in M(\mathcal{P}_i)$* passes *if*

$$\mathrm{TV}\big(\hat{p}_i(\cdot \mid C, a), \operatorname{marg}_i m(C)(a)\big) \leq \eta + w\big(n_i(C, a)\big) \qquad \text{for every } (C, a) \text{ with } n_i(C, a) \geq \bar{n}, \tag{5}$$

where TV *is total-variation distance. The* active set $\widehat{M}_i(\mathcal{P}_i, r_i)$ *collects the passing theories. Under a committed profile, call m η-*adequate *if* $\mathrm{TV}(p_i^*(\cdot \mid C, a), \operatorname{marg}_i m(C)(a)) \leq \eta$ *at every visited pair and* η-inadequate *if the inequality fails strictly at some visited pair.*

Lemma 5.2 (Test consistency)**.** *Fix a committed profile played in all generations and suppose no theory's distance to an on-path conditional equals η exactly. As generations accumulate, almost surely every η-adequate theory is eventually always in the active set and every η-inadequate theory is eventually never in it.*

Proof. Visited pairs accumulate observations without bound almost surely under i.i.d. full-support states, so $\hat{p}_i(\cdot \mid C, a) \to p_i^*(\cdot \mid C, a)$ by the strong law of large numbers. The triangle inequality places $\mathrm{TV}(\hat{p}_i, \operatorname{marg}_i m(C)(a))$ within any $\epsilon > 0$ of $\mathrm{TV}(p_i^*, \operatorname{marg}_i m(C)(a))$ from some generation on; the genericity hypothesis converts strict inequalities into eventual acceptance and rejection, and the slack sequence absorbs finite-sample fluctuation before that generation. □

The deterministic case is the degenerate instance $\eta = 0$, $\bar{n} = 1$, $w \equiv 0$: passing is exact consistency with every retained episode.

Judged updating. Beliefs carry two tiers. The *reference posterior* $\bar{\mu}_i$ begins at a full-support prior on $M(\mathcal{P}_i)$ and updates by likelihoods alone; it is never conditioned on the test. The *acting belief* μ_i is the reference posterior restricted and renormalized to the current active set, so that belief never rides on a rejected theory; when the active set is empty, the acting belief is undefined and the carry-forward clause of assumption 3.4 governs. Because restriction always applies to the unrestricted tier, the acting belief is well defined whenever the active set is nonempty—including when a fluctuating test empties the set and later refills it—provided the reference posterior gives the refilled set positive mass. In the stochastic case we therefore maintain that every marginal $\operatorname{marg}_i \theta(a)$ has full support, which keeps all likelihoods positive; in the deterministic case no assumption is needed, since rejection coincides with a zero-likelihood event, refill cannot occur, and judged updating reduces to ordinary Bayesian conditioning. After a refinement, both tiers restart from the full-support bridge prior on the expanded space (section 8). Judged updating is the formal junction between the test and the posterior—between judgment and understanding (remark 4.2)—and it guarantees, in section 9, that surviving acting beliefs are supported on adequate theories.

5.2 Dissonance and exposure

Definition 5.3 (Causal dissonance)**.** *Agent i experiences* causal dissonance *at the end of a generation if $\widehat{M}_i(\mathcal{P}_i, r_i) = \varnothing$: every expressible theory is rejected. The expressible set does not disappear; no element of it accounts for the record. The immediate question has a determinate defect but not a determinate answer:* what unrepresented difference among the pooled states explains the heterogeneity of their consequences?

Dissonance is the model's rendering of a question for intelligence. It is localized (the failing cells are identified by the test), it is directed (an adequate repair must reconcile the record), and it is semantically open (no expressible object names the missing distinction). Whether dissonance ever occurs, however, depends on whether play makes the falsity of the language visible.

Definition 5.4 (Exposure and concealment)**.** *Under a committed profile, awareness $\mathcal{P}_i$ is* exposed *if every $m \in M(\mathcal{P}_i)$ is η-inadequate. A committed profile is* concealing *for agent i if some expressible theory is η-adequate under it while false as a claim about τ at some state its cells cover.*

Proposition 5.5 (Exposure and dissonance)**.** *Under a fixed committed profile, agent i experiences dissonance from some generation onward almost surely if and only if $\mathcal{P}_i$ is exposed.*

Proof. Immediate from lemma 5.2: exposure means every expressible theory is eventually rejected; its failure means some theory is eventually always accepted. □

Exposure has two sources, and the distinction organizes much of what follows. Heterogeneity inside a cell surfaces either because *other agents play differently across the pooled states*—strategic differentiation inside one's categories, which changes the visited profiles and the induced mixtures—or because *the environment punishes uniform play across causally different states in an observable way*. Both reappear as the corrective forces in section 9.

Example 1 (continued). Let $\beta = 1$ and suppose the firms' priors over the constant theories are heterogeneous: firm 1 considers θ^0 more likely than θ^1, and firm 2 the reverse. A direct computation shows that, against any conjecture about the rival, offering architecture 0 is subjectively strictly dominant for firm 1 and architecture 1 for firm 2 whenever $\mu_i(m^a) > (2-\beta)\,\mu_i(m^{a'})$; at $\beta = 1$ this is simple belief asymmetry. The firms therefore commit to differentiated constant policies. Consider a generation in which s_{00} and s_{11} are served. With differentiated actions, the winner-take-all payoffs reveal the objectively preferred architecture at each state: firm 1 wins at s_{00} (payoffs $1, 0$) and loses at s_{11} (payoffs $0, 1$). No constant assignment is consistent with both episodes—m^0 fails at s_{11}, m^1 at s_{00}, and m^n predicts one-half at both—so each firm's active set empties: causal dissonance, in its exact deterministic form. Had the firms instead pooled on the same architecture at both states, each would have received one-half at every episode, an outcome consistent with all three constant theories: the pooled profile is concealing, and coarse awareness would have survived its own evidence indefinitely. Belief heterogeneity, by inducing differentiated play, is what made the shared language's failure visible.

6 Inquiry: restoration, direction, and fallibility

Dissonance poses a question the current language cannot answer. This section characterizes what a successful answer must accomplish, shows that the requirement directs inquiry without determining its output, and establishes that any inquiry process meeting the paper's disciplines must be fallible.

6.1 Restorative refinements

For $\mathcal{P}, \mathcal{P}' \in \Pi(S)$, write $\mathcal{P} \prec \mathcal{P}'$ when $\mathcal{P}'$ strictly refines $\mathcal{P}$. After a refinement, the retained record is recoded (assumption 3.3) and the test is re-applied with the recoded counts.

Definition 6.1 (Restorative refinement)**.** *$\mathcal{P}'$ is* restorative *at $(\mathcal{P}_i, r_i)$ if $\mathcal{P}_i \prec \mathcal{P}'$ and $\widehat{M}_i(\mathcal{P}', r_i) \neq \varnothing$ under the recoded record; it is* minimally restorative *if no restorative $\mathcal{P}''$ satisfies $\mathcal{P}_i \prec \mathcal{P}'' \prec \mathcal{P}'$. Write $\mathcal{R}(\mathcal{P}_i, r_i)$ for the set of minimally restorative refinements.*

Proposition 6.2 (Cellwise coherence)**.** *Call a cell of a refinement* coherent *if some single mechanism passes the test on the cell's recoded observations. Then $\widehat{M}_i(\mathcal{P}', r_i) \neq \varnothing$ if and only if every cell of $\mathcal{P}'$ is coherent.*

Proof. Theories factor across cells—$M(\mathcal{P}') = \Theta^{\mathcal{P}'}$, and the constraints in equation (5) apply cell by cell—so a passing assignment exists exactly when a passing mechanism can be chosen for each cell independently. □

Coherence is the general form of a conflict structure. Token sets are *mutually incoherent* when no single mechanism accommodates their union; every restorative refinement must separate mutually incoherent sets, since a cell containing both would be incoherent. In the deterministic running example incoherence is a binary relation between labeled episodes and restoration is a graph-coloring problem; in general, a mixture of three mechanisms may defeat every single mechanism even though each pair of tokens is individually accommodatable, so restoration is a hypergraph coloring. Two further structural facts matter. First, minimal restorations, when they exist, are typically multiple—as the example shows vividly. Second, existence itself is guaranteed only in the deterministic case.

Remark 6.3 (Irreparable records)**.** *With deterministic mechanisms, restoration is always possible: the singleton partition recodes each token to its own state, where the generating mechanism fits exactly.*

With stochastic mechanisms this guarantee fails. An unlucky finite sample at a single state can produce an empirical conditional outside the tolerance of every *mechanism, and no partition of S can separate observations generated at the same state; then $\mathcal{R}(\mathcal{P}_i, r_i) = \varnothing$ and the record is* irreparable*: dissonance names a defect that no expressible refinement can repair. Because $\mathcal{R}$ is computable from the record, the agent recognizes irreparability; the technology fails surely (definition 6.4); and the participation rule of section 7 assigns the question no value, so no inquiry cost is wasted on it. Under recurring visitation, irreparability is transient—empirical conditionals converge to the generating mechanism's, which lies within any positive tolerance—but a record whose critical pairs stop being visited can remain irreparable forever, a possibility the dynamics of section 9 classifies within dissonant stasis.*

6.2 The technology

Definition 6.4 (Representation-producing technology)**.** *A representation-producing technology for agent i is a possibly randomized map ϕ_i from $(\mathcal{P}_i, r_i)$ to $\{\mathcal{P}_i\} \cup \mathcal{R}(\mathcal{P}_i, r_i)$: it fails, returning the current partition, or succeeds, producing a minimally restorative refinement. When $\mathcal{R}(\mathcal{P}_i, r_i) = \varnothing$, the range is the singleton $\{\mathcal{P}_i\}$ and the technology fails surely. It is* truth-blind*: its output distribution depends only on $(\mathcal{P}_i, r_i)$, never directly on τ. A complexity ordering κ on refinements—the analyst's model of which conceptual products the agent's comparison capacities can generate—may restrict the range further; κ is a primitive of the technology, not an object of the agent's optimization.*

The technology is the paper's stated representational primitive. No formal model derives a semantically new concept from nothing; the contribution is to show how experience *directs* a limited, fallible generative capacity and why that capacity does not amount to a prior over future theories. The output, and even the set of possible outputs, is not a pre-insight subjective menu: the agent cannot name a candidate refinement, assign it a probability, or condition a policy on it before production succeeds (section 3.5).

6.3 Nonidentification and the price of robustness

Theorem 6.5 (Observational nonidentification)**.** *Let $\tau \neq \tau'$ be objective assignments inducing the same distribution over agent i's records under the committed profiles played. Then any truth-blind technology, and any truth-blind inquiry-and-formulation rule, has the same output distribution in both worlds. If no single theory is η-adequate under both worlds at the visited pairs, no truth-blind rule selects an adequate formulation with probability one in both.*

Proof. The rule's input is the record, whose distribution is identical by hypothesis, so its output distribution is identical. If the adequate sets are disjoint, any realized output is inadequate in at least one world with the corresponding probability. □

Proposition 6.6 (Robust sufficiency)**.** *A partition sufficient for every assignment in a candidate set $\mathcal{T}$ must refine $\bigvee_{\tau' \in \mathcal{T}} \mathcal{P}^{\tau'}$, the join of the candidates' causal-equivalence partitions. A representation* *guaranteed to accommodate all observationally equivalent worlds is accordingly finer than any single world requires; a technology that economizes on distinctions must accept fallibility.*

Proof. Sufficiency for τ' is refinement of $\mathcal{P}^{\tau'}$ (definition 3.2); refinement of each candidate's partition is refinement of their join. $\square$

Example 1 (continued). After the dissonant generation, the record contains two labeled episodes: architecture 0 objectively preferred at s_{00}, architecture 1 at s_{11}. Of the fifteen partitions of four states, exactly four are minimally restorative—each must separate s_{00} from s_{11}: the row partition $\{\{s_{00}, s_{01}\}, \{s_{10}, s_{11}\}\}$, the column partition $\{\{s_{00}, s_{10}\}, \{s_{01}, s_{11}\}\}$, and the two partitions isolating one labeled state. The record directs inquiry (eleven partitions are excluded) without determining it (four remain). A natural complexity ordering counts the coordinates needed to describe a refinement: rows and columns are one-coordinate distinctions, the isolating partitions require two. A factor-seeking technology thus produces the row or the column refinement—and here direction runs out. Define the *column world* $\tau'(s_{rc}) = \theta^c$: it agrees with the row world at both labeled states, so the two worlds generate identical records (theorem 6.5 in exact form). No truth-blind rule can formulate correctly in both, a randomized rule succeeds in the worst case with probability at most one-half, and by proposition 6.6 a partition sufficient for both worlds must be the full partition into singletons: robustness costs every distinction the language can draw, which is exactly what a minimal technology economizes on. The example also locates the failure mode of each residual candidate: whichever factor refinement arrives, its unique consistent theory extrapolates the wrong prediction to the two unlabeled states in one of the two worlds.

7 The economics of inquiry

Whether to prosecute a question must itself be a decision, and it must be priced at exactly the moment ordinary decision theory is silent: at dissonance the posterior is undefined (definition 5.3), and the candidate answers are not represented objects (definition 6.4). This section shows that an honest valuation nevertheless exists: a statistic computed entirely from retained experience that is invariant across every representation successful inquiry could produce.

Assumption 7.1 (Costly, elective, dissonance-triggered production)**.** *The technology ϕ_i operates only when agent i experiences dissonance and elects inquiry, $\varepsilon_i = 1$, at cost $\gamma_i > 0$ in payoff units. Agent i holds a delabeled success weight $\lambda_i \in (0, 1]$: a subjective probability that an elected episode produces a usable refinement, defined on the event of success alone and not on the identities of possible outputs. On failure or abstention the committed policy is carried forward. Awareness is cumulative: acquired distinctions are never lost, though theories formulated on them may later be rejected.*

The success weight is awareness of unawareness in delabeled form—the agent can believe that inquiry may yield a usable distinction without being able to describe any candidate distinction—a construct with precedent in beliefs about unpredictable novelty (Schipper, 2025; Karni and Vierø, 2017).

Definition 7.2 (The pinned structure). *Let $\mathcal{R}(\mathcal{P}_i, r_i) \neq \varnothing$. Each restorative refinement induces, through recoding, a partition of the retained tokens, and after production only rules constant on that partition's cells are implementable. The* pinned structure $\Lambda_i(r_i)$ *is the finest common coarsening—the meet—of the induced token partitions: a token rule is implementable* whichever repair arrives *exactly when it is measurable with respect to $\Lambda_i(r_i)$, and the separations Λ_i preserves*—pinned distinctions—*are those the record forces on every repair. For a token t, let*

$$\Theta_i(t) = \left\{ m'(C'_{\mathcal{P}'}(t)) : \mathcal{P}' \in \mathcal{R}(\mathcal{P}_i, r_i),\ m' \in \widehat{M}_i(\mathcal{P}', r_i) \right\}$$

collect the mechanisms that some admissible repair assigns to t's recurrence; it is nonempty because restorative active sets are nonempty.

Definition 7.3 (Retained-record leverage). *Under the* subjective recurrence premise—*the agent values a prospective repair over one anticipated recurrence of its retained record, token for token—the leverage of inquiry at a dissonant record with $\mathcal{R}(\mathcal{P}_i, r_i) \neq \varnothing$ is*

$$B_i(r_i) = \max_{\psi} \sum_{t \in r_i} \min_{a_{-i} \in A_{-i}} \min_{\theta \in \Theta_i(t)} \Big(\mathbb{E}_{x \sim \theta(\psi(t), a_{-i})}\big[u_i(x_i)\big] - \mathbb{E}_{x \sim \theta(\sigma_i(t), a_{-i})}\big[u_i(x_i)\big] \Big), \tag{6}$$

the maximum taken over $\Lambda_i(r_i)$-measurable rules ψ from tokens to own actions, where $\sigma_i(t)$ denotes the action the carried-forward policy takes at t's awareness cell. Set $B_i(r_i) = 0$ when there is no dissonance or when $\mathcal{R}(\mathcal{P}_i, r_i) = \varnothing$.

Proposition 7.4 (Answer invariance and the participation rule). *$B_i(r_i)$ is well defined, computable from the deliberative record together with the known structure (Θ, u_i), and invariant across the elements of $\mathcal{R}(\mathcal{P}_i, r_i)$: every Λ_i-measurable rule is implementable after any successful production, and the maximand of equation (6) makes no reference to which repair arrives. Moreover $B_i(r_i) \geq 0$. Electing inquiry has represented value $\lambda_i B_i(r_i) - \gamma_i$, so agent i elects if and only if $\lambda_i B_i(r_i) \geq \gamma_i$; in particular, elections occur only at dissonant records that admit some repair. Finally, $B_i(r_i) > 0$ if and only if some pinned rule strictly and robustly improves on the carried-forward policy: questions whose every admissible answer supports current behavior are worthless at any positive cost, however much model multiplicity remains.*

Proof. Each $\mathcal{P}' \in \mathcal{R}(\mathcal{P}_i, r_i)$ induces a partition of the retained tokens, and a Λ_i-measurable rule is constant on the cells of every such partition, since each induced partition refines their meet; by assumption 3.3 the rule can therefore be implemented as a $\mathcal{P}'$-measurable policy at recurrences, whichever $\mathcal{P}'$ arrives. Every object in equation (6) is a function of the record, Θ, and u_i. The carried-forward policy is itself Λ_i-measurable—each induced token partition refines the partition of tokens by current awareness cells, hence so does their meet—so taking $\psi = \sigma_i$ shows $B_i \geq 0$, and $B_i > 0$ exactly when some admissible rule achieves a strictly positive robust gain. With probability λ_i production succeeds and the pinned gain is secured; the extrapolated component of any full formulation is assigned no represented value at the election; on failure or abstention the carried-forward policy continues. When $B_i = 0$—including whenever there is no dissonance or the record is irreparable (remark 6.3)—the strict cost $\gamma_i > 0$ makes abstention optimal. $\square$

The participation rule is a coherent non-Bayesian criterion operating precisely where Bayes is undefined, and its variables respect every discipline of section 3.5: the leverage quantifies over retained tokens and their recorded outcomes—"situations like the episodes I lost will recur, and whatever distinction inquiry yields would let me act on them"—never over candidate representations. Two comparative properties follow immediately and organize the applications. *Action separation, not information*: the value of a question tracks whether its resolution could change behavior, not how much residual uncertainty it would resolve; a maximally uncertain agent whose every admissible repair prescribes its current actions rationally declines to inquire. *Rational cessation*: an agent whose record contains no evidence against its current actions has $B_i = 0$, so sustained success extinguishes inquiry even though awareness remains incomplete.

Remark 7.5 (Where exactness and opacity live in the formula)**.** *Two features of equation (6) deserve comment. First, normalization: the valuation horizon is one full anticipated recurrence of the retained record, so B_i grows as evidence accumulates; any fixed-horizon normalization—per episode, per generation—rescales B_i and γ_i together and changes nothing. Second, the inner minimum over $\Theta_i(t)$ is where* counterfactual transparency *lives. When consequences identify forgone values—as in winner-take-all markets, where a lost episode's record pins the mechanism every repair must assign it—the minimum is over a singleton and the bound is exact; the laboratory's leverage of one-half is this case. When consequences conceal forgone values, $\Theta_i(t)$ is large, the minimum drags each term toward zero, and B_i understates the objective value of repair: opaque environments rationally suppress inquiry. How much an environment lets agents price their own questions is itself a dimension of its diagnosticity, alongside exposure.*

Example 1 (continued). At the dissonant record, both firms' committed constant policies disagree with exactly one revealed label (firm 1 lost at s_{11}, firm 2 at s_{00}). Every restorative refinement separates the two labeled tokens, and implementing a token's revealed label at its recurrence gains exactly one-half against *every* rival action: if the rival plays the winning architecture, the switch converts a differentiated loss (0) into a pooled half; if the rival switches, it converts a pooled half into a differentiated win (1). Hence $B_i = \frac{1}{2}$ for both firms—identical leverage from opposite experiences—and each elects inquiry if and only if $\lambda_i/2 \geq \gamma_i$. The valuation used no conjecture about the rival and no reference to which of the four admissible refinements might arrive. In the notation of equation (6): the pinned structure separates the two labeled tokens, every repair's active theory assigns each its revealed mechanism ($\Theta_i(t)$ is a singleton—the transparency of winner-take-all payoffs), and the mechanism and rival minima are degenerate, so the bound is exact. For rational cessation and its converse, append a validation generation in which firm 1 has acquired the row refinement (certain of the true row assignment) while firm 2 has acquired the column refinement (certain of the false column assignment): their policies differ at the two off-diagonal states, firm 1 wins both, and the labels revealed there falsify firm 2's assignment. Firm 1's record now agrees with its policy everywhere, so $B_1 = 0$: the firm holding the truth rationally never inquires again, and its awareness remains permanently incomplete. Firm 2's record disagrees at two tokens, so $B_2 = 1$: defeat has concentrated the evidence, the dissonance, and the incentive in the losing firm, which elects whenever $\lambda_2 \geq \gamma_2$ and, upon success, is driven stepwise to complete awareness.

8 The pivot: formulation, warrant, and the return to Bayes

Successful production yields a refinement $\mathcal{P}'_i$; *formulation* selects a theory $m' \in \widehat{M}_i(\mathcal{P}'_i, r_i)$ on the recoded record; and the agent forms a full-support *bridge prior* $\mu_i^+ \in \Delta(M(\mathcal{P}'_i))$, from which ordinary updating resumes. Nothing in Bayesian coherence determines μ_i^+ from the pre-dissonance belief—the domains differ—and this nonuniqueness is not a defect to be repaired but the boundary between insight and judgment. Renormalizing μ_i^+ on the active set yields what we call *representation-relative coherence*: the confidence the new language awards its own theories. The question of this section is how much of that confidence the evidence actually licenses.

Definition 8.1 (Reflective assessment, warrant, and the gap)**.** *A reflective assessment problem is a pair $(\mathcal{T}, \nu)$: a finite set of candidate objective assignments and a full-support reflective prior. Given the record and a successful production-and-formulation event, the* selection-adjusted warrant *W of $(\mathcal{P}'_i, m')$ is the ν-posterior probability that m' is objectively correct—that $\tau'(s) = m'(C_{\mathcal{P}'_i}(s))$ for all s—conditional on the record* and *on the production event. The* warrant gap *G is the representation-relative coherence probability of m' minus W.*

Theorem 8.2 (The insight is not evidence)**.** *If the technology is truth-blind, conditioning on the production event adds nothing to the record: W equals the ordinary reflective posterior of m''s correctness given the record alone. Post-restoration fit is guaranteed by the restoration property, so passing the test carries no information beyond the record either. Among candidates in $\mathcal{T}$ that induce the same record distribution, prior odds are preserved exactly.*

Proof. The reflective posterior on $\tau' \in \mathcal{T}$ is proportional to $\nu(\tau')$, times the record's likelihood under τ', times the probability of the production event given the record; truth blindness makes the last factor a function of the record alone, constant across $\mathcal{T}$, so it cancels. Fit is a deterministic function of record and output (definition 6.1), hence has likelihood one in every world consistent with the event. The last claim applies the cancellation to candidates with identical record likelihoods. □

The theorem locates the evidential boundary: everything evidential is in the record; nothing evidential is in the arrival of the insight or in its retrospective success at organizing the experience that produced it. The practical force is a prohibition on double counting. The same experience that directed representation production cannot also confirm the resulting theory, because any restorative output fits it by construction. Truth blindness is the polar case; were production correlated with the objective assignment—social transmission from an informed source—the arrival would itself be evidence, and the adjustment could move confidence in either direction.

Corollary 8.3 (Self-audit requires sufficiency for the candidates)**.** *The assessment $(\mathcal{T}, \nu)$ is expressible by agent i only if $\mathcal{P}_i$ is sufficient for every $\tau' \in \mathcal{T}$, hence only if $\mathcal{P}_i$ refines the join of proposition 6.6. An agent holding a minimal restoration therefore generally cannot formulate the comparison that measures its own warrant gap: the overconfidence is invisible from within the representation that produces it. Short of that refinement, warrant must be supplied externally—by fresh evidence whose likelihoods differ across candidates, by an assessor with broader awareness, or by a future self.*

Proof. Entertaining τ' as a hypothesis requires expressing it, which is sufficiency (definition 3.2); sufficiency for every candidate is refinement of the join. □

The pivot is now complete, and it is the paper's fulcrum. Dissonance broke the agent's Bayesian competence; inquiry produced a distinction; formulation and the bridge prior rebuilt a probability space; and judgment—fresh evidence with differential likelihoods, honestly accounted—resumes as ordinary updating on the expanded language. The agent has returned to the Bayesian world carrying a warrant gap of known location and, by corollary 8.3, generally unknowable (to it) size.

Example 1 (continued). Suppose firm i's technology produces the column refinement. The recoded record pins the unique consistent theory: architecture 0 on the first column, 1 on the second. Representation-relative coherence assigns it probability one. Now conduct the reflective assessment with $\mathcal{T} = \{\text{row world}, \text{column world}\}$ and a symmetric prior. The two worlds generate identical records (theorem 6.5), and the technology is truth-blind, so by theorem 8.2 the reflective posterior equals the prior: $W = \frac{1}{2}$, and the warrant gap is exactly one-half. The same gap afflicts the firm that acquires the *row* refinement: its coherence in the true assignment is also probability one, and its warrant is also one-half. Unwarranted certainty is not a symptom of error—early confidence outruns the evidence even when it happens to be right, and what separates the two firms is the objective law, which neither observes. By corollary 8.3, neither firm can conduct this assessment itself: expressing both candidate worlds requires the full partition into singletons. One fresh differentiated observation at an off-diagonal state closes the gap completely—the two worlds predict opposite labels there—while pooled observations, in any number, close nothing. Warrant, like leverage, is purchased with differentiated action.

9 Strategic dynamics: when everyone lives in a Bayesian world

The chain now runs inside a population. Theories determine committed policies; policies determine which conditionals are visited; visited conditionals determine dissonance and leverage; and refinements feed back into theories. This section characterizes where the process comes to rest. The object of interest is not a strategic equilibrium but an epistemic condition: a configuration of the population in which *every agent permanently inhabits a Bayesian world*—models fit, questions are dead, and no represented calculation prices further inquiry above its cost.

Definition 9.1 (Self-confirming unawareness equilibrium)**.** *A profile of epistemic states $\xi_i = (\mathcal{P}_i, r_i, \bar{\mu}_i, q_i, \sigma_i)$, $i \in N$—awareness, retained record, reference posterior, conjecture, and committed policy, with acting beliefs μ_i derived by judged updating—is a* self-confirming unawareness equilibrium *(SCUE) if, under the play it induces,*

(i) each σ_i is subjectively optimal (equation (3));

(ii) each conjecture q_i agrees with the play induced by σ_{-i} at every on-path cell;

(iii) no dissonance occurs: each agent's active set, computed from its record as play accumulates, is almost surely eventually nonempty, and each acting belief μ_i is supported on η-adequate theories; and

(iv) no agent elects inquiry: $\lambda_i B_i(r_i) < \gamma_i$ along the induced records, which holds in particular whenever $B_i = 0$.

The record belongs in the state: dissonance status and leverage are functions of retained evidence, so two histories with identical partitions, beliefs, and policies but different records can differ in equilibrium status.

Conditions (i)–(ii) are subjective rationality with confirmed conjectures: the self-confirming conditions of the fixed-language tradition (Fudenberg and Levine, 1993; Kalai and Lehrer, 1993b). Conditions (iii)–(iv) are the new, epistemic ones: the language survives its own evidence, and the question has no priced value. A SCUE in which every partition is sufficient for τ and beliefs concentrate near the truth is simply a self-confirming equilibrium; the concept's interest lies entirely in the other cases.

Remark 9.2 (The grain of truth, restored endogenously)**.** *Rational-learning convergence requires a grain of truth: the environment absolutely continuous with respect to beliefs (Kalai and Lehrer, 1993a). Under unawareness the condition can fail at the level of language—the truth may be inexpressible—and statistical dissonance is precisely the detector of that failure. Each successful refinement restores the grain* on the path of play*: post-restoration, some expressible theory matches the visited conditionals within tolerance. A SCUE is a configuration in which the grain of truth holds on-path for every agent and no one has a priced reason to look for the off-path remainder.*

Assumption 9.3 (Regularity)**.** *(i)* Generic tolerance. *The tolerance η lies outside the finite set of critical values at which, for some agent, theory, partition profile, and policy profile, the total-variation distance between an induced on-path conditional and the theory's prediction equals η. (ii)* Production randomness. *The randomization in each ϕ_i is independent across elections and of the state and consequence draws, and, conditional on any election—which by proposition 7.4 occurs only at dissonant records admitting a repair, $\mathcal{R}(\mathcal{P}_i, r_i) \neq \varnothing$—production succeeds with probability at least $\underline{\pi} > 0$.*

Part (i) excludes finitely many tolerance values, since agents, theories, partitions, and policies are all finite. Part (ii) is substantive: it rules out technologies that can be elected forever without ever producing (remark 9.6).

Theorem 9.4 (Epistemic absorption)**.** *Under assumption 9.3, almost surely:*

(a) at most $n(|S| - 1)$ refinements occur, and every agent's awareness partition is eventually constant;

(b) only finitely many inquiry elections occur; and

(c) there is an almost surely finite generation after which no agent's language changes and no agent pays an inquiry cost: every agent's epistemic life thereafter is judged-Bayesian updating within a fixed language, punctuated at most by dissonance episodes that fail the participation threshold.

Moreover, on the event that the committed profile is eventually constant,

(d) every agent's active set is eventually constant, every theory it contains is η-adequate under the terminal play at every recurrent observation pair, and each agent is eventually either never dissonant—it lives in a Bayesian world—or in dissonant stasis*: dissonance recurs while $\lambda_i B_i < \gamma_i$, in particular whenever $B_i = 0$, as at irreparable records ($\mathcal{R}(\mathcal{P}_i, r_i) = \varnothing$).*

The proof is in appendix A.

Proposition 9.5 (Rest points and persistence)**.**

(i) If awareness partitions, committed policies, conjectures, and acting beliefs are all eventually constant almost surely, then the limiting configuration, restricted to the agents not in dissonant stasis, is a self-confirming unawareness equilibrium.

(ii) Conversely, consider a configuration whose epistemic states satisfy the SCUE conditions, whose play generates only records with $B_i = 0$, and in which each agent's committed action at each cell is strictly optimal under every conjecture and every theory in its active set.

(a) With deterministic mechanisms, the configuration persists: dissonance, elections, and policy changes never occur, and the SCUE conditions continue to hold, with conjectures converging empirically to induced play.

(b) *With stochastic mechanisms, suppose in addition that every theory in every active set is η'-adequate under the configuration's play for some $\eta' < \eta$, and that the strict optimality extends to every $(\eta + \varepsilon)$-adequate theory for some $\varepsilon \in (0, 2(\eta - \eta')]$. Then for every $\delta > 0$ there is a count threshold $N(\delta)$ such that, if the configuration is reached with at least $N(\delta)$ observations at every recurrent pair, then with probability at least $1 - \delta$ play never changes, dissonance and elections never occur, and the SCUE conditions hold in every subsequent generation, with conjectures converging empirically to induced play.*

The proof is in appendix A.

Remark 9.6 (What the theorem does and does not claim)**.** *Theorem 9.4 is a statement about the epistemic process—the death of inquiry—not about the convergence of play. Within a Bayesian world, committed policies may in principle continue to cycle as beliefs and conjectures chase one another; that is the classical convergence problem of learning in games (Fudenberg and Levine, 1993; Kalai and Lehrer, 1993a), it is orthogonal to this paper's contribution, and the theorem deliberately does not resolve it. Part (d) and the rest-point characterization are accordingly conditional on play stabilizing, which in every configuration featured in this paper is immediate: their beliefs make one action per cell strictly dominant. Separately, if the success bound in assumption 9.3(ii) is dropped, one further absorbing pattern becomes possible: perpetually elected, perpetually failing inquiry—a question worth asking that is never answered, whose economics (paying γ_i forever for a repair that never arrives) merit separate study.*

Remark 9.7 (Rational anomaly tolerance)**.** *Zero-leverage dissonant stasis is a recognizable condition, not a pathology of the model: the agent knows its causal account is broken—every expressible theory is rejected—yet the break disturbs no represented action comparison, so repairing it is worth nothing. Standing anomalies tolerated by working scientists and managers have exactly this structure, and the model locates them: anomaly tolerance is rational precisely when pinned distinctions carry no robust action leverage. A sharper instance is the irreparable record (remark 6.3): there $\mathcal{R} = \varnothing$, no expressible repair exists at any price, and the anomaly is not merely unprofitable to fix but impossible to fix within the current grammar.*

Proposition 9.8 (On-path adequacy, and where error lives)**.** *In a SCUE, every theory in the support of every μ_i is η-adequate at every visited pair. All surviving falsity is confined to what the configuration does not expose: mechanism assignments at states pooled with causally different states whose mixture imitates a single mechanism at the visited profiles, and counterfactual claims at unvisited profiles.*

Proof. Immediate from definition 9.1(iii) and lemma 5.2; the residual clause is the complement of the visited pairs' constraints. □

Proposition 9.9 (Persistent performance differences)**.** *There exist primitives and absorbing configurations in which agents hold different terminal partitions and theories, expected per-generation payoffs differ strictly and permanently, and no disadvantaged agent has a priced repair: for each such agent either $B_i = 0$ or $\lambda_i B_i < \gamma_i$.*

The proof is by the instances below, and the mechanism deserves emphasis. Performance differences persist not because an equilibrium concept freezes them but because the process that would erase them has died economically: the disadvantaged configuration generates no dissonance, or generates no leverage, or prices its question below the cost of asking. Terminal heterogeneity in awareness is path dependence in the strong sense—agents with identical preferences and identical Bayesian competence, facing the same objective system, end at different languages and different payoffs because their histories posed different questions.

Definition 9.10 (Diagnostic environments)**.** *The primitives* (Θ, τ, ρ, u) *are* diagnostic *if, under every profile sustainable in a SCUE, every expressible theory that is payoff-relevantly false—false at some state where the truth would change its holder's optimal action—is* η*-inadequate.*

Proposition 9.11 (No payoff-relevant error in diagnostic environments)**.** *In a diagnostic environment, every SCUE is free of payoff-relevant error. Payoff-relevant falsity survives only in environments that admit concealing profiles, and it requires them to be sustained: collective error needs the absence of* both *corrective forces, strategic variety inside agents' categories and environmental punishment of uniform misservice.*

Proof. By definition 9.1(iii), theories surviving in a SCUE are not η-inadequate; in a diagnostic environment every payoff-relevantly false theory is. The second claim restates definition 5.4: a false theory survives only under concealing profiles, and proposition 5.5 makes concealment necessary as well as sufficient. □

Example 1 (continued). Set $\beta = 1$ and suppose both firms acquired the column refinement after the discovery generation and became certain of the false column assignment. Each then plays the column policy; the firms pool at every state; every payoff is one-half; and one-half is exactly what the column theory predicts under pooled play. The configuration is a SCUE with payoff-relevant error: policies are subjectively optimal, conjectures confirmed, no dissonance ever occurs (the profile is concealing), and $B_i = 0$ for both firms—so the configuration survives even a free, infallible inquiry technology. The warrant gap of section 8 is permanent: unbounded experience accumulates while each firm remains exactly half-warranted, the missing half being the reflective weight on the world their shared language cannot express. Now let $\beta < 1$. In the first generation containing an off-diagonal state, the pooled firms receive $\beta/2$ where their theory predicts one-half: the demand side reveals the label, dissonance returns, leverage turns positive, and the reflective posterior becomes degenerate. Payoff-relevant self-confirmation is a knife-edge property of perfectly forgiving demand—the $\beta < 1$ environments are diagnostic—and the destroyed value $1 - \beta$ at each misserved episode is itself the corrective signal: what demand withholds from the firms, their epistemics receives as evidence. Finally, persistence: with $\beta = 1$, let firm 1 hold the row refinement and the true assignment while firm 2 remains coarse with $\gamma_2 > \lambda_2 B_2$. Firm 1 wins both off-diagonal states and splits the rest, earning three-quarters to firm 2's one-quarter, forever; firm 2 sits in dissonant stasis, its question priced below its cost.

Every route out—strategic variety, demand harshness, cheaper inquiry—is exactly one of the model's named forces.

10 Discussion

10.1 Implications for the theory-based view

The model supplies the theory-based view with the object its formal treatments presuppose and, in doing so, changes what several of the view's claims mean.

Strategic variety has a pre-probabilistic source. In the model, agents with identical preferences, identical Bayesian competence, and identical objective opportunities diverge because their experiences posed different questions, their technologies produced different repairs, and their positions extinguished inquiry at different points. Heterogeneous prior understanding generates heterogeneous questions; questions generate heterogeneous languages; languages generate heterogeneous theories, plans, and payoffs. This sequence begins before probability does, and it is path-dependent in the strong sense: the order of experiences shapes the terminal language, and agents can converge to the same behavior while retaining permanently different representations (proposition 9.9).

The value of experimentation is not information. A question is worth asking when some admissible answer would change action (proposition 7.4), not when its resolution would be most informative. This reframes theory testing: high-entropy uncertainty with no action leverage is rationally ignored, and narrow contrasts that straddle an action threshold are rationally pursued. It also explains rational cessation—why successful firms stop asking—and the loser's advantage in inquiry: competitive defeat concentrates the evidence, the dissonance, and the incentive in the defeated firm.

Early confidence outruns evidence even when correct. The warrant results discipline a claim at the center of the empirical program on scientific entrepreneurship (Camuffo et al., 2020; Zellweger and Zenger, 2023). A founder's theory that beautifully organizes the experiences that generated it has, by theorem 8.2, earned nothing by that organization: the fit was guaranteed by construction. Validation requires evidence whose likelihood differs across the candidate worlds—and, by corollary 8.3, the founder typically cannot even express the comparison, which is a formal argument for outside challenge (boards, investors, advisors with different awareness) as a warrant technology rather than a monitoring technology.

Markets discipline theories through two channels, and their absence is predictable. Payoff-relevant collective error requires both concealment by play and forgiveness by demand (proposition 9.11). Industries where customers absorb misservice—captive buyers, no substitutes, third parties bearing the cost—can sustain shared false theories indefinitely; industries with strong outside options discipline theories even under consensus. Where demand is forgiving, the only corrective is strategic variety in beliefs, which gives belief heterogeneity a social value distinct from its private value: the entrant with a deviant theory is the industry's epistemic infrastructure.

Anomaly tolerance is an equilibrium phenomenon. Firms that know their causal account is broken and rationally decline to fix it (remark 9.7) are not failing at learning; their standing anomalies carry no action leverage. The model predicts where tolerated anomalies accumulate: in transparent-consequence environments, near decisions whose robust optimum is insensitive to the unresolved structure.

10.2 The cognitional interpretation

The model's architecture tracks a classical account of inquiry (Lonergan, 1992; Ryall and Wilkins, 2018), and the correspondence is disciplining rather than decorative: it dictates which formal objects must not be conflated. Experience is the flow of episodes; the question for intelligence is dissonance, which localizes a defect without naming its repair; heuristic anticipation is restoration, minimality, and the complexity ordering, which constrain an answer without supplying its content; insight's formal footprint is the refinement; formulation is the mechanism assignment on the new cells; reflective judgment is the consistency test together with the warrant accounting, faculties formally distinct from the posterior (remark 4.2); and decision is the committed policy, which then determines what experience will teach next. The election ε_i adds the economic layer: an unrestricted desire to know, prosecuted under restricted resources. Nothing in the mathematics represents the conscious act of insight itself; the model represents its products and its price, which is what an economic theory can responsibly claim.

10.3 Limitations and the research program

Four boundaries of the present model define the program's next steps. First, the mechanism vocabulary Θ is fixed: agents lack distinctions, never concepts of consequence-generation. Expanding the vocabulary itself—deeper unawareness—is the natural second paper in the sequence, and the awareness-transition machinery here is built to receive it. Second, the model prices participation in inquiry but not its intensity or duration, and the technology's success is exogenous; endogenizing the depth of inquiry connects to models of costly deliberation and planning (Ryall, 2026), where cached prior conclusions govern when reconsideration is rational—the theory-to-plan transition. Third, agents are individuals; the theory-bearing *firm*—how insights become shared languages, collective intentions, and organizational commitments—is a separate paper resting on the same primitives (Ryall and Wilkins, 2018). Fourth, the competitive consequences here run through payoffs in a fixed game; joining the front end developed here to the cooperative back end of value capture (Gans and Ryall, 2017; Bryan et al., 2022), where awareness itself earns rents and disclosure is strategic, completes the arc. Within the present model, the flagged analytical items are general conditions for the convergence of play within Bayesian worlds (remark 9.6), equilibrium in the anti-coordination region that value destruction creates, counterfactual transparency as a design variable, and Markov state processes.

11 Conclusion

This paper models the step the theory-based view has described but not formalized: the origin of a strategic theory. The model draws a formal boundary between learning within a causal language and producing a new one, and it prices the crossing. Bayesian updating cannot expand the language and cannot even detect that expansion is needed; detection is statistical dissonance, a faculty distinct from the posterior. Dissonance directs repair without determining it; observational equivalence makes any economical inquiry technology fallible; and the value of a question is computable, invariantly across its inconceivable answers, at exactly the moment the posterior is undefined. The return to Bayesian life is disciplined by a warrant account whose central theorem is easily stated: the insight is not evidence. And because theories cause the actions that cause the evidence, inquiry is almost surely absorbed: eventually everyone lives in a Bayesian world—in configurations that preserve false theories, heterogeneous languages, and permanent performance differences exactly when neither strategic variety nor environmental harshness disturbs them. Strategy's oldest question—why do similar firms perform differently?—here receives an answer that begins before beliefs: because they came to different questions, and some questions died before they were asked.

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A Proofs of Theorem 9.4 and Proposition 9.5

Probability framework. All randomness is carried by three mutually independent i.i.d. families on a common probability space: state draws $\{s_{g,t}\}$ over generations g and episodes $t \leq T$, with law ρ; consequence randomizers $\{U_{g,t}\}$, uniform on $[0,1]$, from which $x_{g,t}$ is generated through a fixed measurable representation of the kernel $\tau(s_{g,t})(a_{g,t})$; and production randomizers $\{V_{i,g}\}$, uniform on $[0,1]$ (assumption 9.3(ii)). Let $\mathcal{F}_g$ denote the σ-algebra generated by all randomness through generation g. Committed policies and election indicators for generation g are $\mathcal{F}_{g-1}$-measurable, since they are functions of records, beliefs, and conjectures formed from earlier generations; conditional on $\mathcal{F}_{g-1}$ and on an election by agent i, production succeeds with probability at least $\underline{\pi}$.

Part (a). Each successful production strictly refines one agent's partition, and, by cumulative awareness (assumption 7.1), no partition ever coarsens. The total cell count $\sum_i |\mathcal{P}_i|$ is integer valued, starts at no less than n, never exceeds $n|S|$, and strictly increases at every success. Hence at most $n(|S|-1)$ successes occur on every path, and each agent's partition is eventually constant.

Part (b). Enumerate elections in order of occurrence across agents and generations, and let $Z_k \in \{0,1\}$ indicate success of the k-th election. By proposition 7.4, every election occurs at a dissonant record with $\mathcal{R}(\mathcal{P}_i, r_i) \neq \varnothing$, so assumption 9.3(ii) applies: on the event that at least k elections occur, the conditional probability of $Z_k = 1$ given the information available at the k-th election is at least $\underline{\pi}$. By Lévy's extension of the Borel–Cantelli lemma (Williams, 1991, Thm. 12.15), on the event that infinitely many elections occur, $\sum_k Z_k = \infty$ almost surely. But $\sum_k Z_k$ counts successful productions, which part (a) bounds by $n(|S|-1)$. Hence the event that infinitely many elections occur has probability zero.

Part (c). Let G^* be the larger of the last refinement generation and the last election generation, almost surely finite by (a) and (b). After G^*, no language changes and no inquiry cost is paid. Because the election rule is mechanical—agent i elects exactly when it is dissonant and $\lambda_i B_i \geq \gamma_i$—every dissonance episode after G^* must fail the participation threshold. All other epistemic activity after G^* is judged-Bayesian updating within the fixed terminal language.

Part (d). Condition on the event that the committed profile is eventually constant, equal to σ^* from some generation $G_2 \geq G^*$, and fix an agent i with terminal partition $\mathcal{P}_i^\infty$. Call an observation pair (C, a) *recurrent* if $S_{\sigma^*}(C, a) \neq \varnothing$ and *frozen* otherwise. A frozen pair is visited at most finitely often, so its count, its empirical conditional, and the test constraint it imposes on any theory are constant from some generation on. A recurrent pair is visited in each episode with probability $\rho(S_{\sigma^*}(C, a)) > 0$, independently across episodes, so its count diverges almost surely; the post-G_2 observations at the pair are i.i.d. with law $p_i^*(\cdot \mid C, a)$ computed under σ^*, and the pre-G_2 observations form a finite prefix, so the cumulative empirical conditional converges to $p_i^*(\cdot \mid C, a)$ almost surely by the strong law applied coordinatewise on the finite space X_i. Fix a theory m and a recurrent pair. If $\mathrm{TV}(p_i^*, \mathrm{marg}_i\, m(C)(a)) < \eta$—equality being excluded by assumption 9.3(i)—then eventually $\mathrm{TV}(\hat{p}_i, \mathrm{marg}_i\, m(C)(a)) < \eta \le \eta + w(n)$ and the constraint is permanently satisfied; if the distance exceeds η, then, since $w(n) \to 0$, the constraint eventually fails permanently. With finitely many theories and pairs, the active set is eventually constant, equal to the set A_i^* of theories satisfying every frozen constraint and lying within η of the terminal on-path conditional at every recurrent pair; in particular, every element of A_i^* is η-adequate under terminal play at every recurrent pair. If $A_i^* \neq \varnothing$, agent i is eventually never dissonant. If $A_i^* = \varnothing$, it is eventually always dissonant; since no election occurs after G^* and the election rule is mechanical, $\lambda_i B_i < \gamma_i$ at every such generation, which includes the case $B_i = 0$ and, in particular, irreparable records with $\mathcal{R}(\mathcal{P}_i, r_i) = \varnothing$. □

Proof of Proposition 9.5(i). Suppose awareness, policies, conjectures, and beliefs are eventually constant almost surely, with limits $(\mathcal{P}_i, \mu_i, q_i, \sigma_i)$. Condition (i) of definition 9.1 holds because each generation's committed policy solves equation (3) at the then-current belief and conjecture, which are eventually the limits. For condition (ii), empirical conjectures converge almost surely to the frequencies induced by the eventually-constant profile, and an eventually-constant conjecture equals its limit, so q_i agrees with induced play at every on-path cell. For condition (iii), an agent not in dissonant stasis is eventually never dissonant by theorem 9.4(d); judged updating keeps the support of the acting belief μ_i inside the active set, which is eventually A_i^*, whose members are η-adequate at every recurrent pair. For condition (iv), elections cease by theorem 9.4(b), so whenever dissonance occurs the participation threshold fails; absent dissonance, $B_i = 0$ by definition 7.3. □

Proof of Proposition 9.5(ii-a). With deterministic mechanisms, condition (iii) of definition 9.1 forces every on-path observation to match the active theories exactly: an active theory assigns each visited (C, a) a degenerate prediction, and a single disagreeing observation would reject it permanently, contradicting almost-sure absence of dissonance. Hence every state in each visited $S(C, a)$ produces the predicted consequence, empirical conditionals are constant, and active sets never change. Dissonance never occurs; $B_i = 0$ at every generated record by hypothesis, so no election and no refinement ever occurs; and strict optimality of the committed action under every conjecture and every active theory fixes σ_i regardless of how the reference posterior and the empirical conjectures evolve. The SCUE conditions therefore continue to hold, with empirical conjectures converging to the constant induced play. □

Proof of Proposition 9.5(ii-b). Fix $\varepsilon \in (0, 2(\eta - \eta')]$ and choose N large enough that $w(n) \leq \varepsilon/2$ for all $n \geq N$ and that

$$\sum_{n \geq N} K\, e^{-n\varepsilon^2/2} \;\leq\; \frac{\delta}{P},$$

where K is the constant in the finite-alphabet Hoeffding bound $\Pr\big(\mathrm{TV}(\hat{p}_n, p^*) > \varepsilon/2\big) \leq Ke^{-n\varepsilon^2/2}$ (a union of Hoeffding bounds over the events of the finite space X_i) and P is the number of recurrent pairs across agents. On the complement of the event that some recurrent pair's empirical conditional ever leaves the $\varepsilon/2$-ball around its on-path conditional after count N—an event of probability at most δ—the following hold in every subsequent generation. Every η'-adequate theory passes the test at every recurrent pair, since $\mathrm{TV}(\hat{p}, \text{prediction}) \leq \varepsilon/2 + \eta' \leq \eta$; frozen constraints are unchanged and already satisfied by the active theories; so active sets remain nonempty and dissonance never occurs. Every theory the test admits is $(\eta + \varepsilon)$-adequate, since $\mathrm{TV}(\text{prediction}, p^*) \leq \eta + w(n) + \varepsilon/2 \leq \eta + \varepsilon$; so the extended strict-optimality hypothesis fixes each agent's committed action at every cell, for every conjecture and every acting belief the judged update can produce. Play therefore never changes; the generated records have $B_i = 0$ by hypothesis, so no election and no refinement ever occurs; and the SCUE conditions hold in every subsequent generation, with empirical conjectures converging to the constant induced play. □

B Supporting computations for the running example

Dominance thresholds. At an awareness cell C, write $p^a = \mu_i(\{m : m(C) = \theta^a\})$ for $a \in \{0,1\}$. Against a rival playing 0, the subjective expected-payoff difference between own actions 0 and 1 is $\frac{1}{2}p^0 - (1 - \frac{\beta}{2})p^1$; against a rival playing 1 it is $(1 - \frac{\beta}{2})p^0 - \frac{1}{2}p^1$; mixtures follow by linearity. Action 0 is strictly dominant on C if and only if $p^0 > (2 - \beta)p^1$, and symmetrically. Certainty in an architecture label makes the labeled action strictly dominant for every $\beta \in [0,1]$, which is what sustains the committed policies in the trap configuration. At $\beta = 1$ the threshold is simple belief asymmetry, which is what differentiates the firms' opening policies.

Unit leverage. Let a retained labeled episode carry revealed label a^* while the committed policy prescribes $a \neq a^*$, and compare the committed policy with one implementing a^* at that recurrence, at $\beta = 1$. If the rival plays a^*: committed yields 0 (differentiated loss), implementing yields $\frac{1}{2}$ (pool). If the rival plays a: committed yields $\frac{1}{2}$ (pool), implementing yields 1 (differentiated win). The gain is exactly $\frac{1}{2}$ in both cases, hence against every rival mixture. Episodes whose labels agree with the committed policy contribute zero. With one disagreeing episode per firm after the discovery generation, $B_i = \frac{1}{2}$; after the validation generation of section 7, $B_1 = 0$ and $B_2 = 1$.

The trap and its knife edge. With both firms certain of the column assignment, the column policy pools at every state; every payoff is $\frac{1}{2}$; and under pooled play every mechanism in $\{\theta^0, \theta^1, \theta^n\}$ consistent with the column labels predicts $\frac{1}{2}$ at $\beta = 1$. No episode is labeled, active sets never empty, and $B_i = 0$; conditions (i)–(iv) of definition 9.1 hold. For $\beta < 1$, at an off-diagonal state the pooled action is objectively wrong, realized payoffs are $\beta/2$ against a prediction of $\frac{1}{2}$, and the payoff level itself reveals the label (payoff $\frac{1}{2}$ implies the pooled architecture is preferred; $\beta/2$ implies the other is). The revealed label conflicts with a diagonal label inside one column cell, the active set empties, the episode enters the disagreement record, and the reflective posterior over the row and column worlds becomes degenerate: dissonance, leverage, and warrant closure arrive together.

Persistence payoffs. With firm 1 playing the true row assignment and firm 2 a constant architecture, the firms pool at two states and differentiate at two, firm 1 winning both differentiated states: average payoffs $\frac{3}{4}$ and $\frac{1}{4}$ under uniform ρ. Firm 2's record rejects all three constant theories (it wins nothing differentiated and loses twice), so it sits in dissonance with $B_2 > 0$; stasis obtains whenever $\gamma_2 > \lambda_2 B_2$.